

Universidad Central de Venezuela
Facultad de Ciencias
Escuela de Computación
Cátedra de Aplicaciones con Tecnologías del Internet
Semestre 2026-01

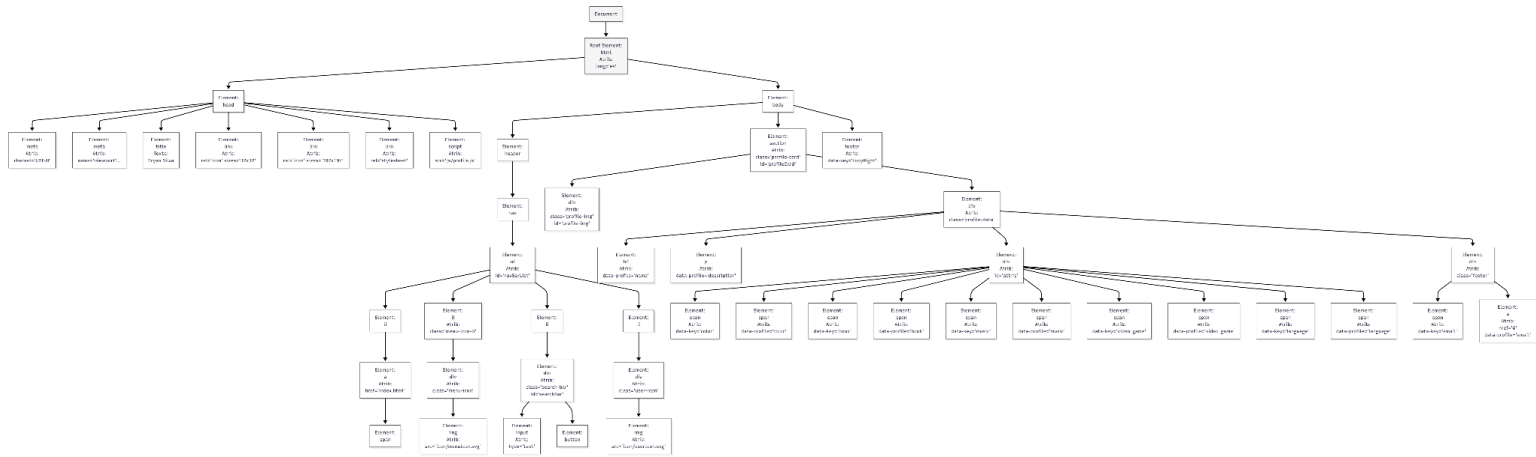
Reto #4

Autor

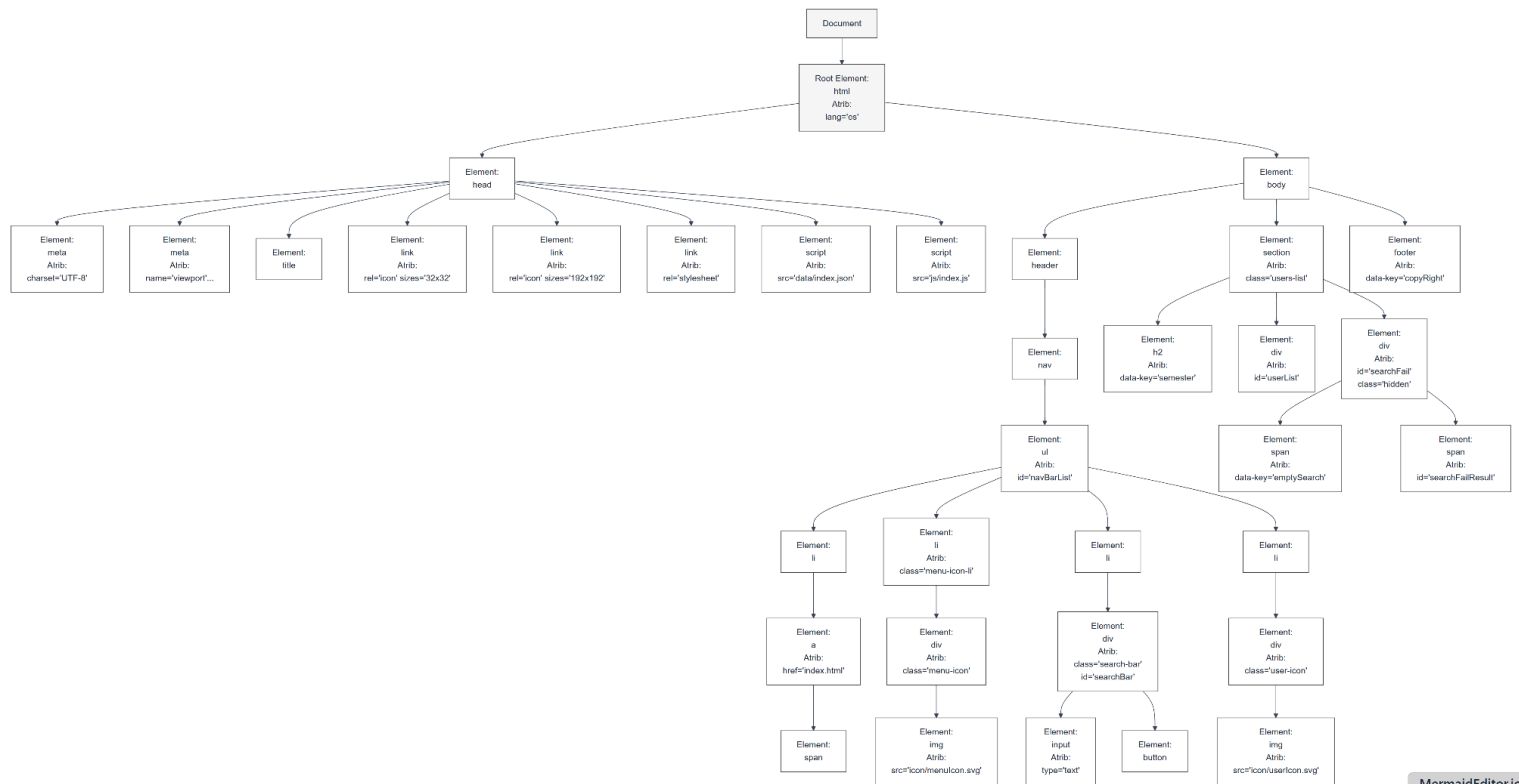
Bryan Silva
27.531.458

Caracas, mayo de 2026

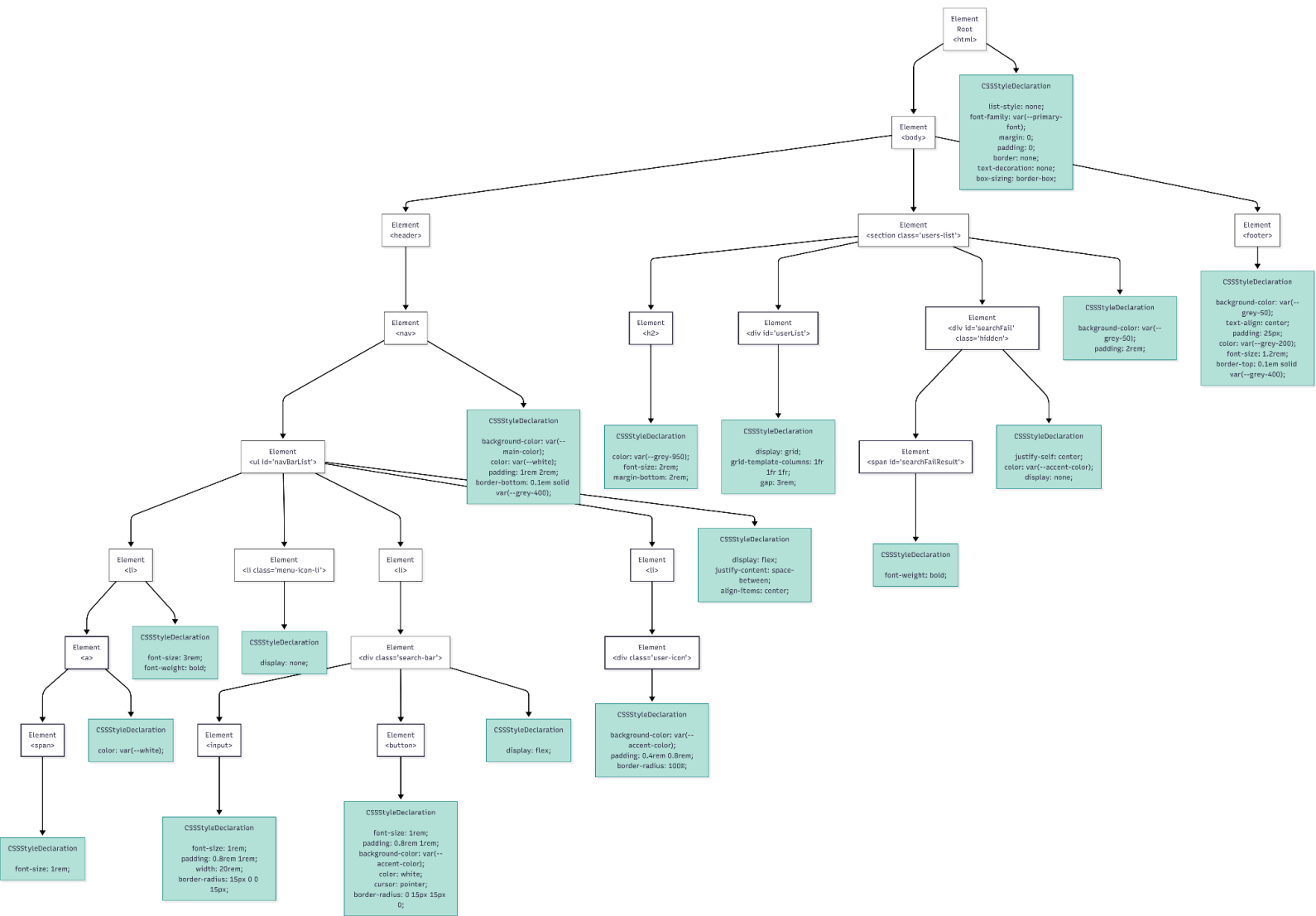
DOM PROFILE



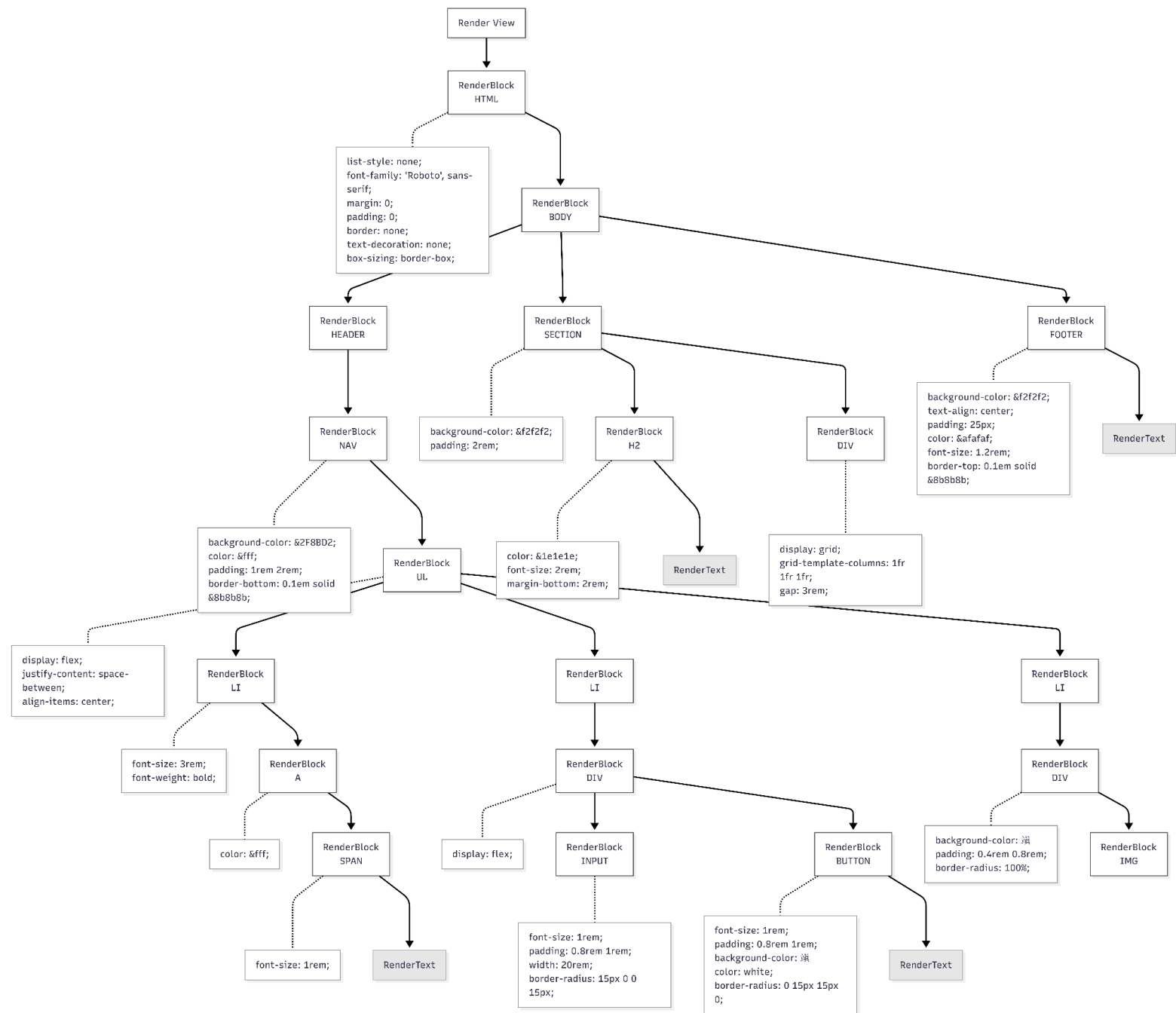
DOM INDEX



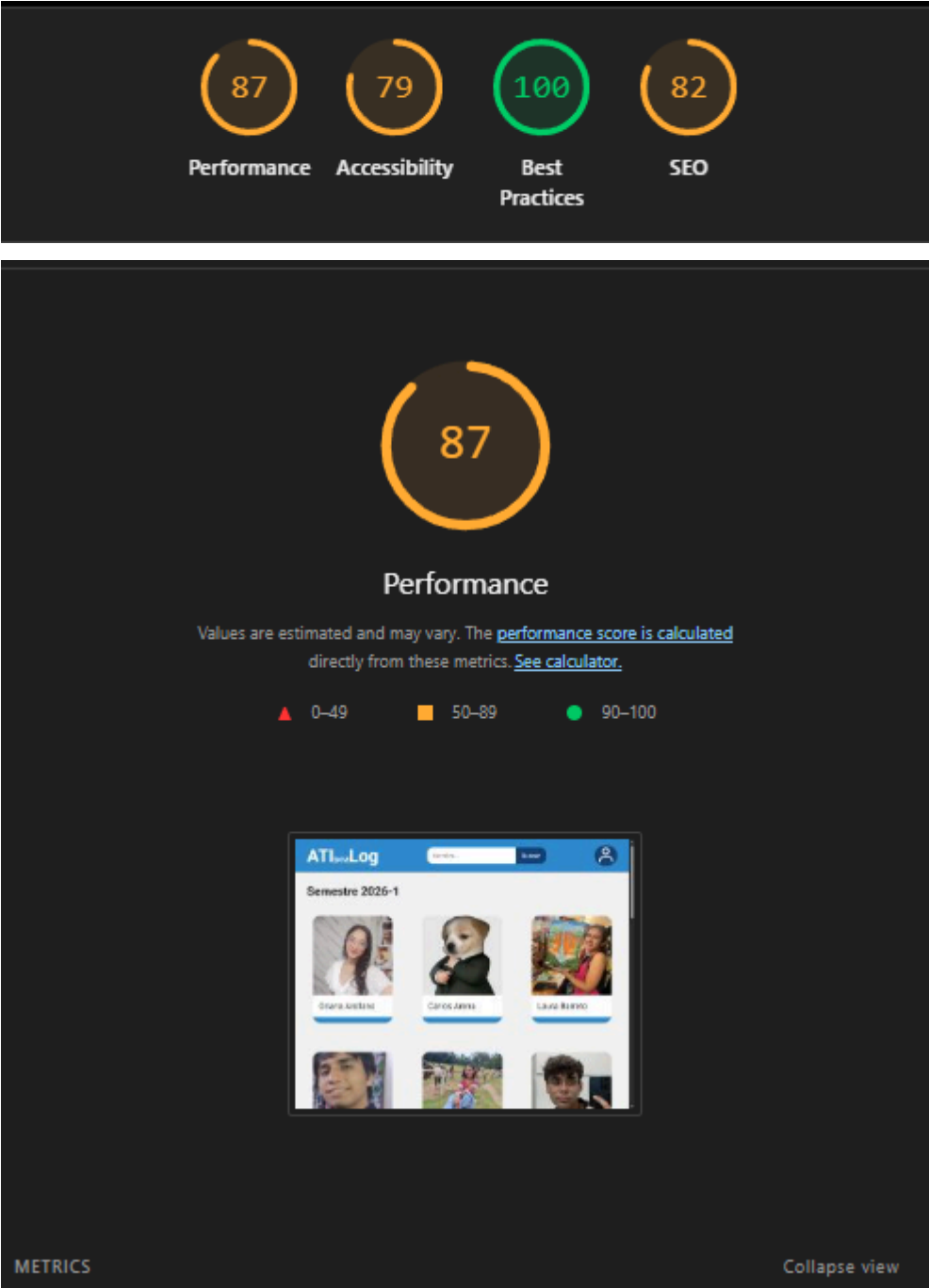
CSSOM INDEX



RENDER TREE DOM



PRUEBAS



METRICS

Collapse view

● First Contentful Paint

0.5 s

First Contentful Paint marks the time at which the first text or image is painted. [Learn more about the First Contentful Paint metric](#)

■ Largest Contentful Paint

2.2 s

Largest Contentful Paint marks the time at which the largest text or image is painted. [Learn more about the Largest Contentful Paint metric](#)

● Total Blocking Time

0 ms

Sum of all time periods between FCP and Time to Interactive, when task length exceeded 50ms, expressed in milliseconds. [Learn more about the Total Blocking Time metric](#)

■ Cumulative Layout Shift

0.103

Cumulative Layout Shift measures the movement of visible elements within the viewport. [Learn more about the Cumulative Layout Shift metric](#)

● Speed Index

0.5 s

Speed Index shows how quickly the contents of a page are visibly populated. [Learn more about the Speed Index metric](#)

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Accessibility

These checks highlight opportunities to [improve the accessibility of your web app](#). Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so [manual testing](#) is also encouraged.

NAMES AND LABELS

▲ Image elements do not have `[alt]` attributes

These are opportunities to improve the semantics of the controls in your application. This may enhance the experience for users of assistive technology, like a screen reader.

CONTRAST

▲ Background and foreground colors do not have a sufficient contrast ratio.

These are opportunities to improve the legibility of your content.

NAVIGATION

▲ Heading elements are not in a sequentially-descending order

These are opportunities to improve keyboard navigation in your application.

BEST PRACTICES

▲ Document does not have a main landmark.

These items highlight common accessibility best practices.

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Best Practices

TRUST AND SAFETY

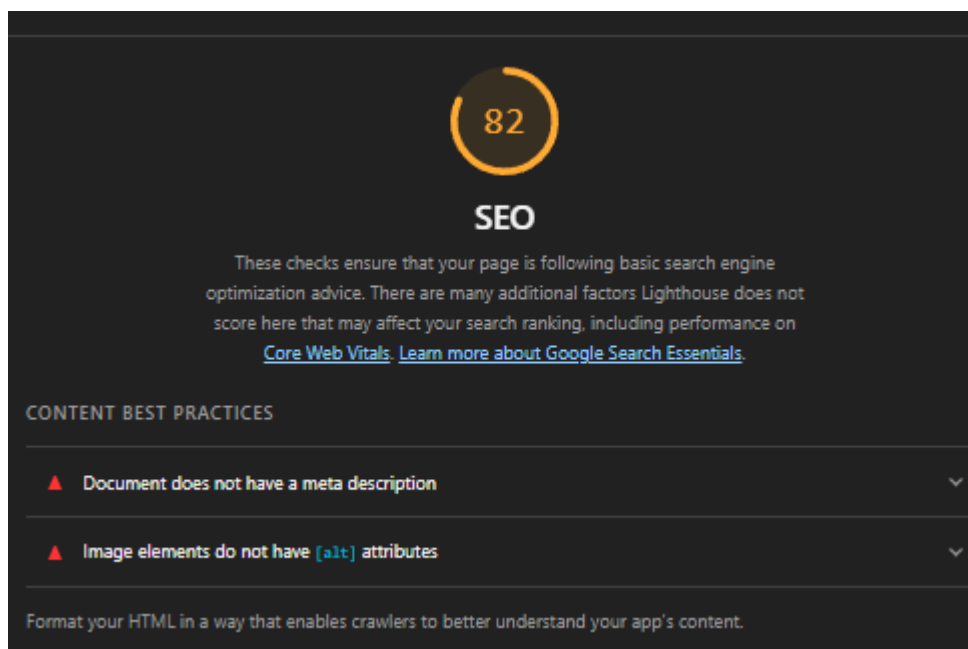
○ Ensure CSP is effective against XSS attacks

○ Use a strong HSTS policy

○ Ensure proper origin isolation with COOP

○ Mitigate clickjacking with XFO or CSP

○ Mitigate DOM-based XSS with Trusted Types



Durante la prueba se usó la herramienta **Lighthouse** integrada en las DevTools de Chromium (Edge específicamente), la cual proporciona una auditoría automatizada y objetiva sobre la calidad general de la aplicación web, evaluando métricas estandarizadas de rendimiento, accesibilidad, buenas prácticas y SEO.

Específicamente, la prueba me permitió visualizar áreas de mejora estructurales en la **Accesibilidad (79/100)** y el **SEO (82/100)** de la página.

La herramienta detectó la ausencia sistemática del atributo alt en las etiquetas de imagen , lo cual penaliza tanto a los lectores de pantalla como a los motores de búsqueda.

Además, evidenció problemas semánticos y de contraste, indicando que el documento carece de una etiqueta de región principal (**<main>**), que los encabezados (**<h2>**, **<h3>**, **etc.**) no descienden en un orden lógico secuencial, y que algunas combinaciones de color entre texto y fondo no cumplen con el ratio mínimo de legibilidad.